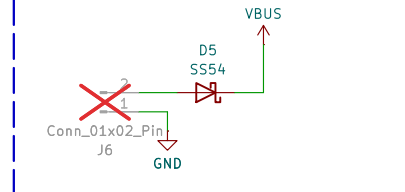
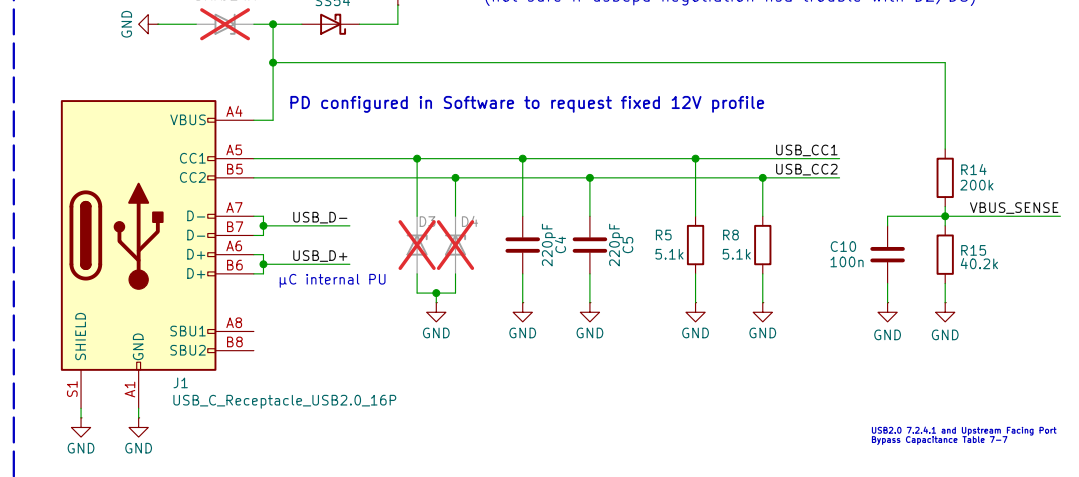


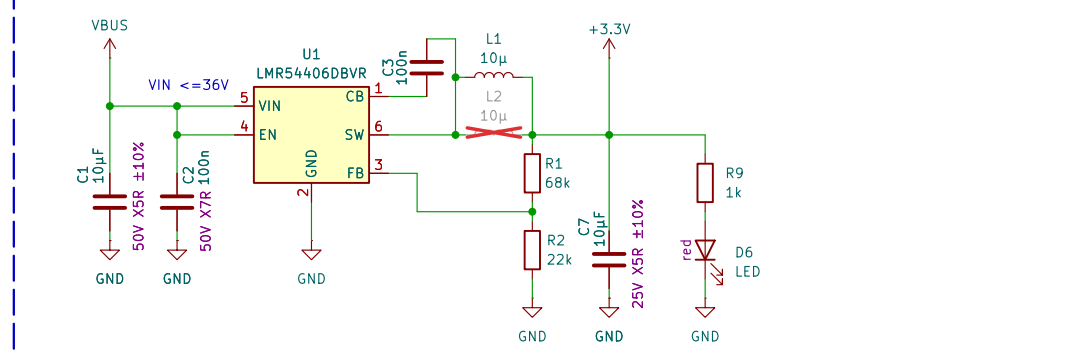
Board Power Supply (+12V)



USB Data onboard $\mu C \rightarrow PC$. USBC Power OPTIONAL
NOT used for first experiments
only if we want to power the Board with 9/12V via USBC-PD later
(not sure if usbcdp negotiation has trouble with D2/D5)



12V $\rightarrow$ 3.3V
-> potentially change to switcher LMR54406BVR
Inductor alternatives: ASP1-40205-100M-T (C3222037), SDCL1V4020-100M-R 74405042100, XAL4040-103ME (C5144883)
handsolder: SPM4020T-100M-LR, ANR4020T100M (C6808085)



μC

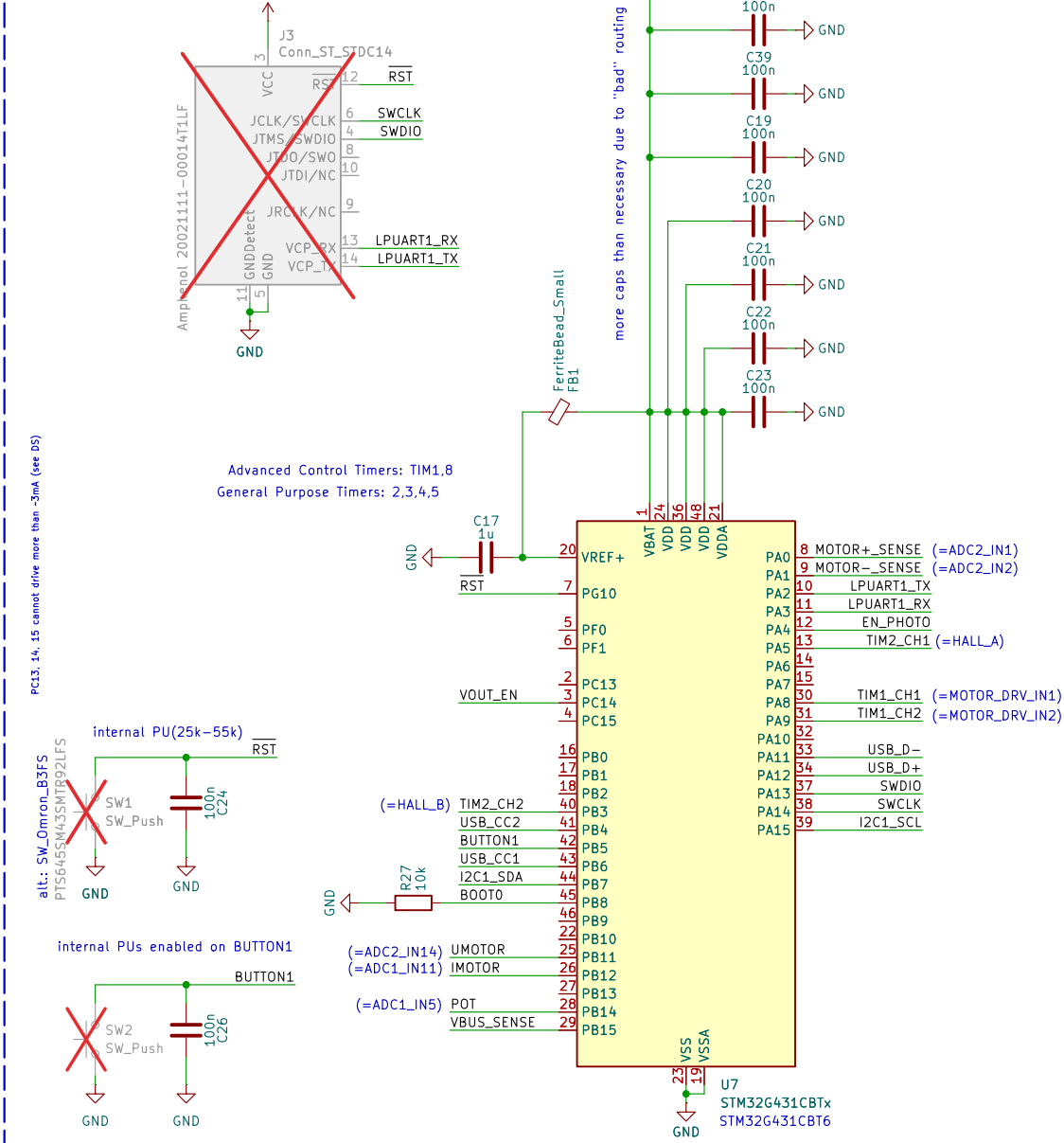
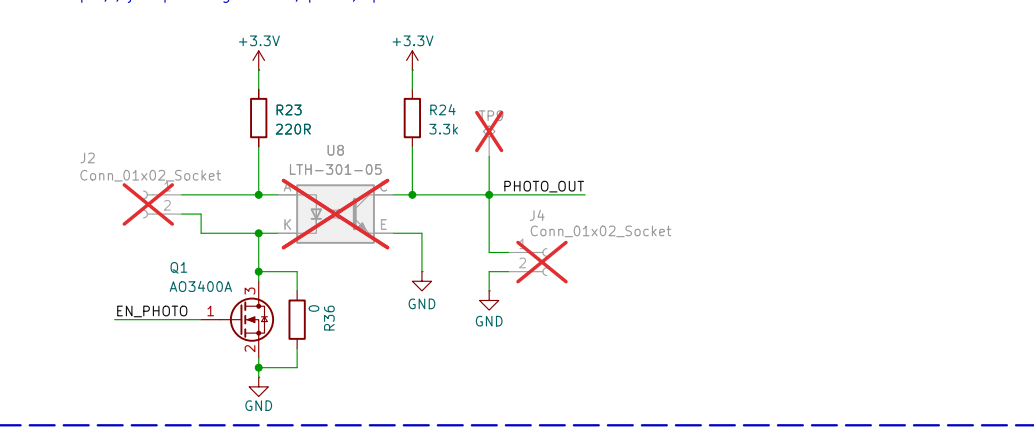


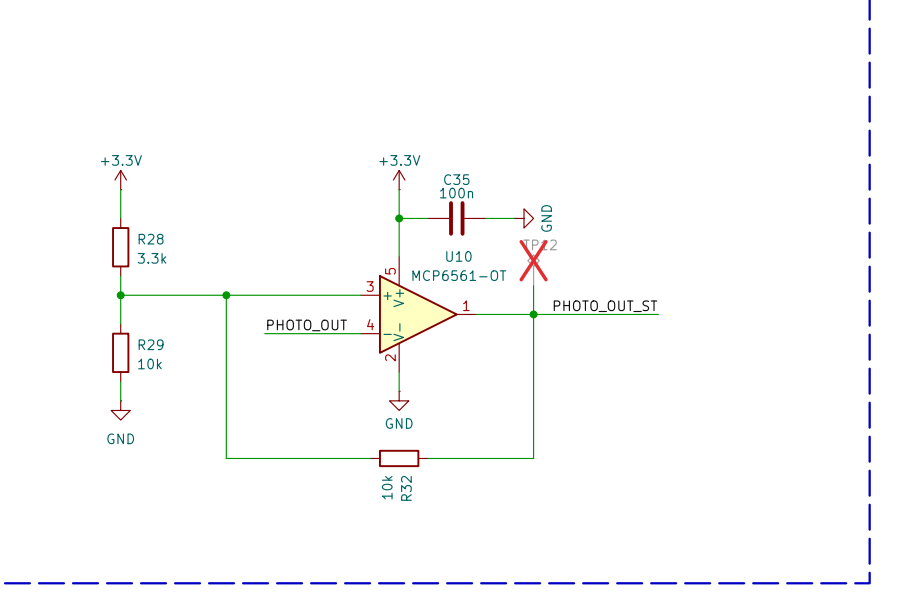
Photo Interruptor

No CTR given in Datasheet
1.)
At $I_F = 20mA \rightarrow$ Datasheet guarantees $> 0.5mA$ photocurrent $\rightarrow$ estimated worst case $CTR \sim 0.5/20 = 2.5\%$
 $\rightarrow$ worst case $I_C @ 10mA$ LED drive $= 240\mu A$
 $\rightarrow$ For $R_{load} = 1k \rightarrow V_{low,out} = 3.3 - 240\mu A * 1k = 3.1V \rightarrow$ only 0.2V swing
2.)
To guarantee saturation in worst case, R_{load} must limit I_C to $240\mu A$
 $R_{load} > (3.3 - V_{CE,sat}) / 240\mu A = 13k\Omega$
3.)
Estimate R_{load} for large enough output swing:
For a swing of 2V $\rightarrow R_{load} = 2 / 240\mu A = 8.3k\Omega$
 $\rightarrow$ we want be rather fast, so try with a smaller R_{load} first
<https://josephehoff.github.io/posts/rpm>

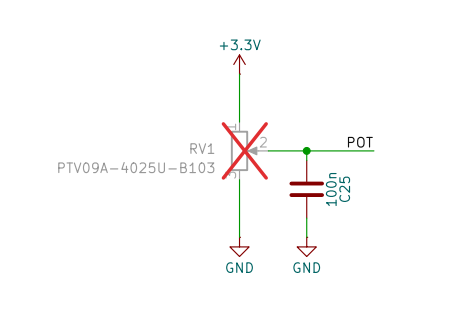


Schmitt-Trigger

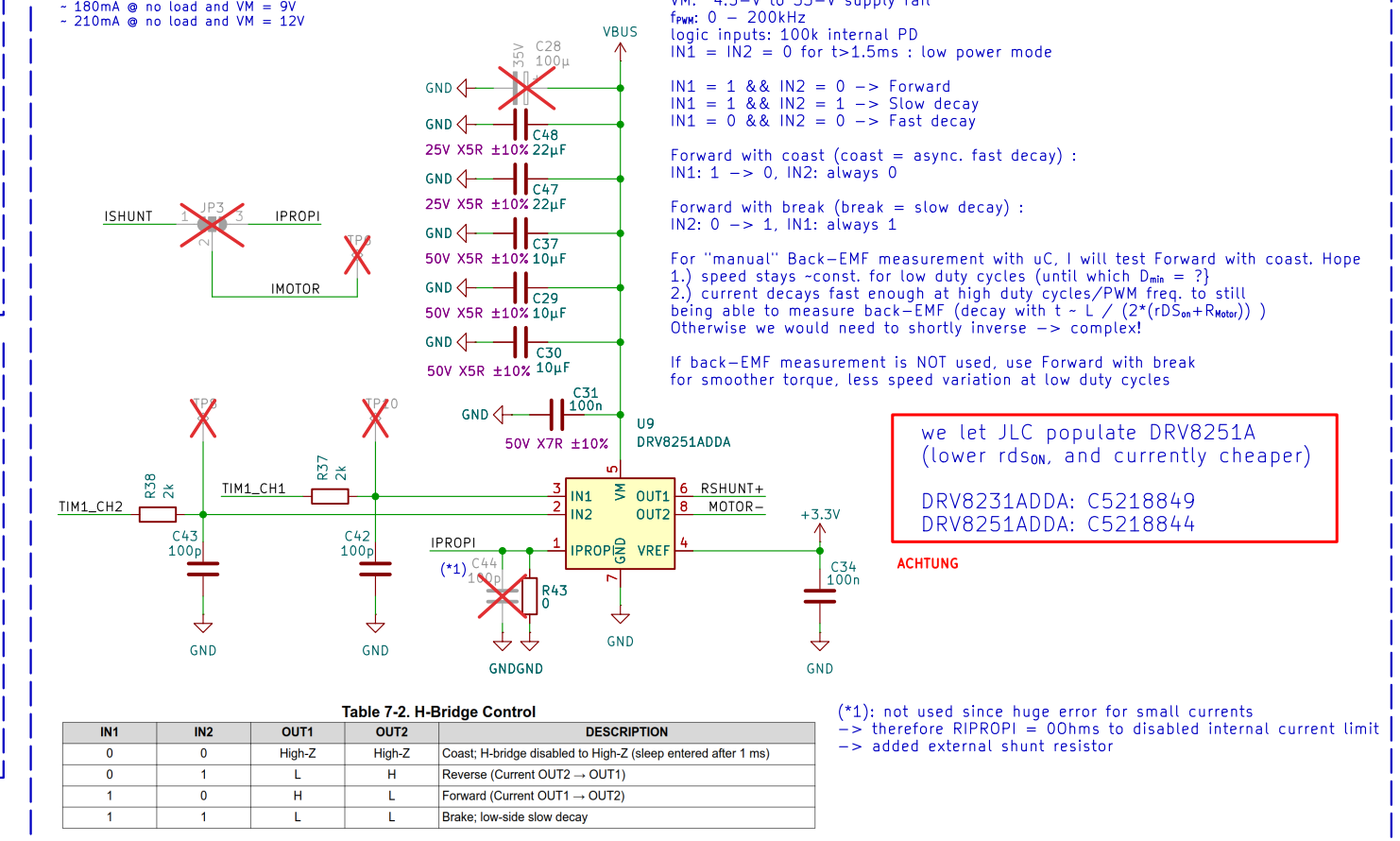
Inverting config
 $\rightarrow$ for Rthevenin $< <$ Feedback : hysteresis-width - independent from threshold center
 $\rightarrow$ source impedance does not influence thresholds
with 1.5V swing from photointerruptor we expect 3.3V (high) $\rightarrow$ 1.8V (low)
 $\rightarrow$ set center threshold to $\sim 2.55V$
 $\rightarrow$ currently: switching threshold: 2.42V, hysteresis: 240mV



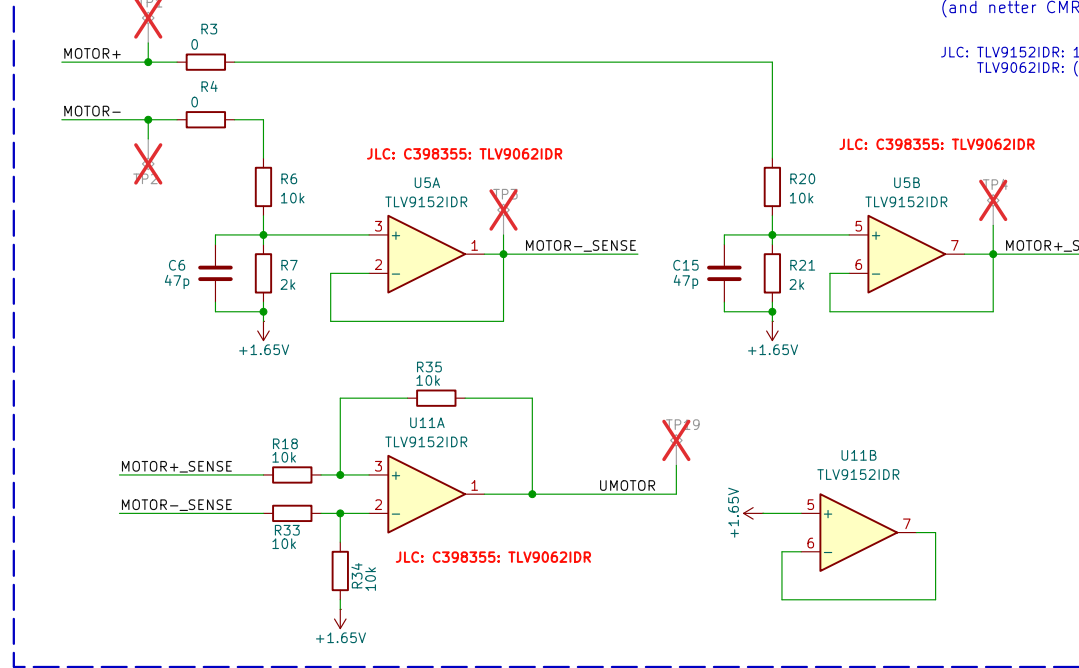
Potentiometer for manual speed control



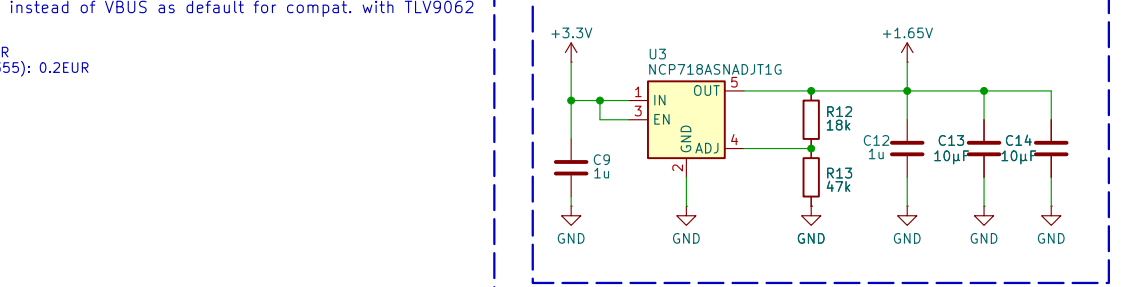
Motor-Driver



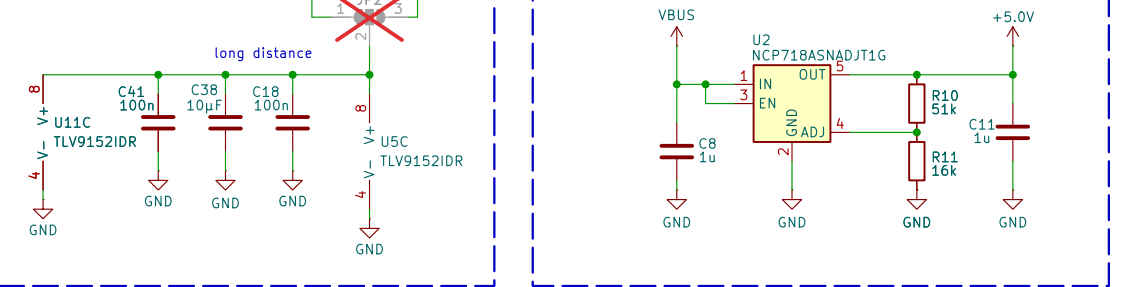
Back-EMF measurement



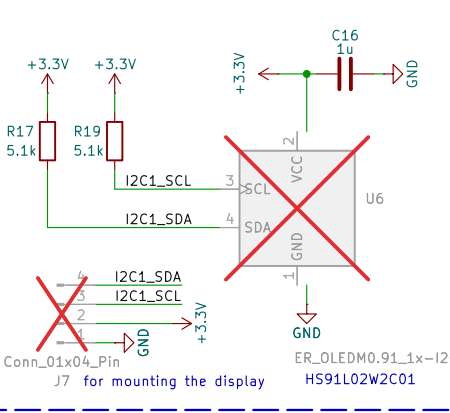
Virtual GND for Back-EMF Measurement



5V for HALL-Supply and Back-EMF Measurement



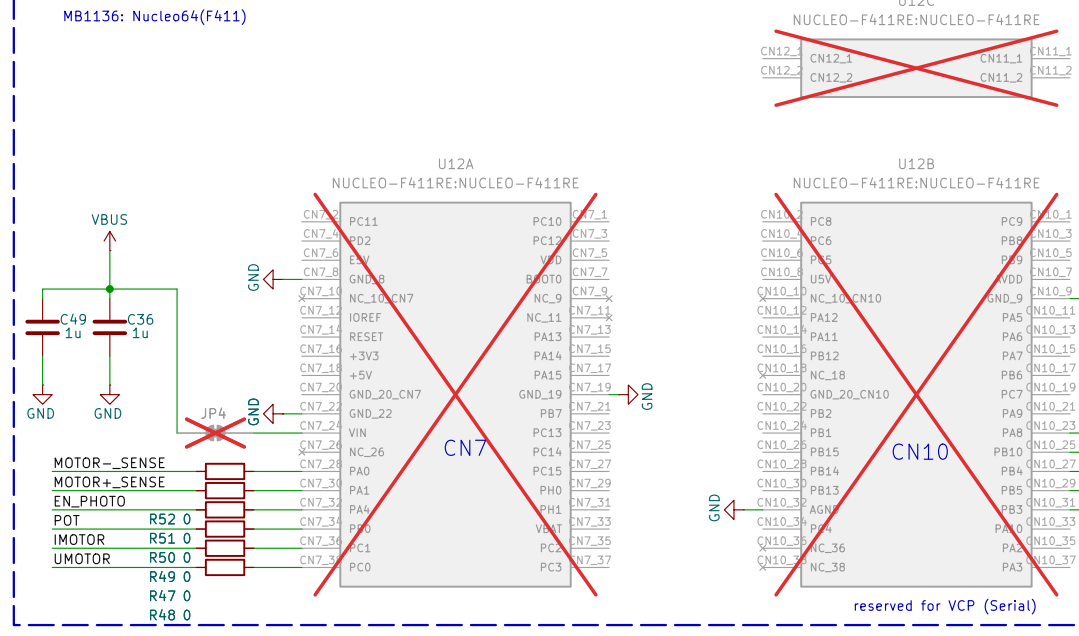
Display (optional)



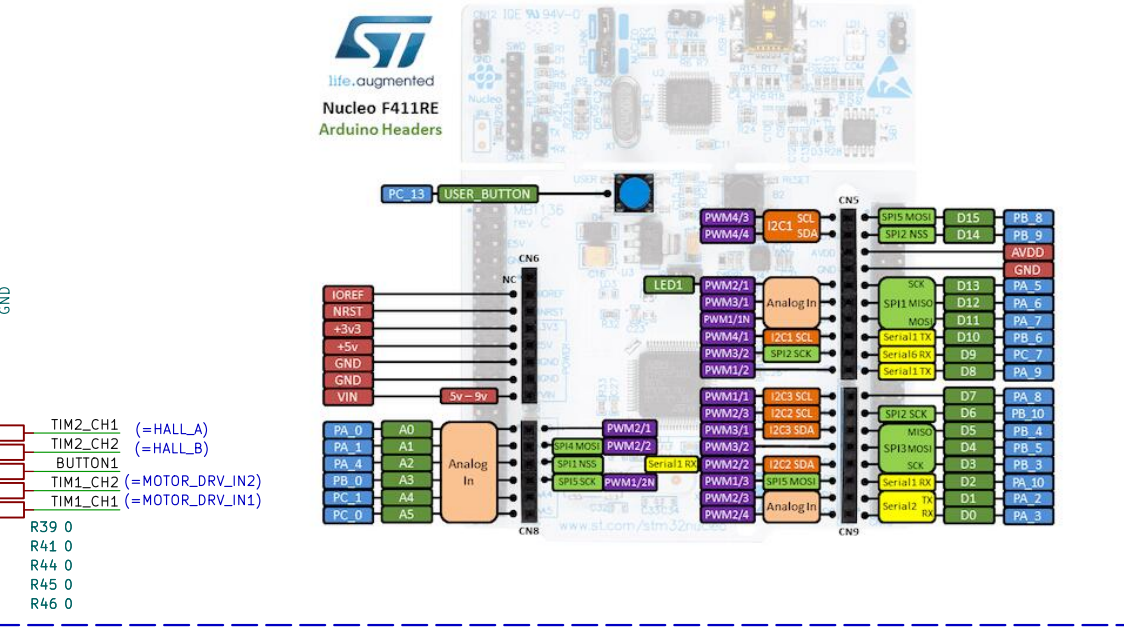
Motor Connector

pins mapped to allow TIM2 to be configured in CubeMX as "combined channels" in Encoder Mode to measure Motor Hall Encoder Signals as quadrature signals

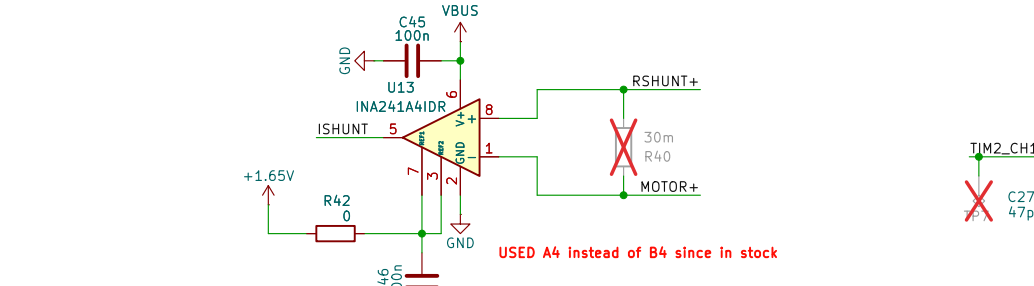
MATLAB



NUCLEO-F411RE:NUCLEO-F411RE



INA241B4IDR: 50IC8 for easy soldering



shift down the 5V to 3V3

